

Multifractal Large Deviations in Quantum Entanglement: Ideal Fermi-Gas Mapping, Markov Tensor Cascades, and Quartic Casimir Symmetry

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Abstract

The thermodynamic formalism of multifractals assigns to any normalised distribution a mass exponent $\tau(q)$ and a singularity spectrum $f(\alpha)$. Applied to the eigenvalues of a reduced density matrix, this formalism turns Rényi entropies into generalised dimensions and the refined Rényi entropy into $f(\alpha_q)$. We use it to compare two notions that are often conflated: the multifractality of single-particle wave functions at localisation transitions, and the multifractal structure of many-body entanglement spectra.

For Gaussian fermionic states we prove that the refined Rényi entropy is exactly the entropy of an ideal Fermi gas at temperature $1/q$ whose single-particle levels are the entanglement energies. The entanglement singularity spectrum is therefore always non-negative and concave. Its dependence on wave-function multifractality is mediated entirely by the density of entanglement energies, which is controlled by inter-eigenstate overlaps rather than by single-state moments.

For tensor-network cascades with correlations between layers, modelled by a Markov chain, the mass exponent is set by a Perron root. Its first-order correction is linear in the subleading eigenvalue λ_2 , and it is real-analytic for $\lambda_2 < 1$. A first-order (Maxwell) transition appears only in the reducible limit $\lambda_2 = 1$.

At the quantum Hall transition, Weyl-group invariance makes the anomalous exponents functions of $u = q(1 - q)$. The measured non-parabolicity b_1 is then the coefficient of the u^2 (higher-Casimir) term. This term appears at four loops in the $2 + \varepsilon$ expansion with the same sign as observed, and its non-vanishing excludes theories in which the relevant operators are current-algebra primaries of a WZW model.

Finally, we compute Gauss–Bonnet cosmic branes in $d = 4$. We confirm that the second Taylor coefficient of $\tau(q)$ tracks C_T/a , find that the refined Rényi entropy turns negative at finite q for positive Gauss–Bonnet coupling, and explain why no correspondence between holographic couplings and the quantum Hall coefficient b_1 should be expected.

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1 Introduction

The multifractal formalism [1, 2] describes a normalised measure $\{\mu_i\}$ on cells of size ε through the scaling of its moments, $\sum_i \mu_i^q \sim \varepsilon^{\tau(q)}$, and through the Legendre transform $f(\alpha) = q\alpha - \tau(q)$, $\alpha = \tau'(q)$, which counts the cells with $\mu_i \approx \varepsilon^\alpha$. The formalism is thermodynamic: q plays the role of an inverse temperature, τ of a free energy, α of an energy and f of an entropy.

Two quite different physical objects are called “multifractal” in quantum many-body physics.

1. *Wave-function multifractality.* At Anderson transitions and at the plateau transition of the integer quantum Hall (IQH) effect, the single-particle probability $|\psi(r)|^2$ has moments $\langle |\psi|^{2q} \rangle / \langle |\psi|^2 \rangle^q \sim L^{-\Delta_q}$ with non-linear anomalous exponents Δ_q [3]. The relevant measure lives on real space and is averaged over disorder.
2. *Entanglement-spectrum multifractality.* The eigenvalues $\{\lambda_i\}$ of a reduced density matrix ρ_A form a normalised measure on the spectrum. Its moments are the Rényi entropies, $S_q = \frac{1}{1-q} \ln \text{Tr} \rho_A^q$. For a 1+1-dimensional CFT the full eigenvalue density is the Calabrese–Lefevre distribution [4].

The same formalism applies to both, but the measures are different, and so are their symmetries and positivity properties. Jia, Subramaniam, Gruzberg and Chakravarty showed that the summed single-site entanglement of a single-particle critical state scales as $\alpha_1 \ln L$ [5], a link through one point of the spectrum only. Dong noted that the refined Rényi entropy is a thermodynamic entropy with respect to the modular Hamiltonian [6], which places the entanglement problem squarely within the thermodynamic formalism.

Results. Building on these works, we obtain the following.

1. *Ideal Fermi-gas theorem* (Theorem 1). For any Gaussian fermionic state, the refined Rényi entropy equals the entropy of an ideal Fermi gas at temperature $1/q$ with the entanglement energies as single-particle levels. Consequently $f_{\text{ent}} \geq 0$ is concave and is fixed by the density of entanglement energies. Wave-function multifractality enters only through that density, which depends on overlaps between different eigenstates (Proposition 2.2).
2. *Markov tensor cascades* (Section 3). For layer-to-layer correlations modelled by a Markov chain, $\tau(q)$ is a Perron root. Its correction is linear in λ_2 , not a power λ_2^q , and it is analytic for $\lambda_2 < 1$. Phase transitions require reducibility or quenched averaging.

3. *Casimir structure at the IQH transition* (Section 4). Writing the measured spectrum in the Weyl-invariant variable $u = q(1 - q)$ identifies b_1 with the coefficient of u^2 . This term arises at four loops in the $2 + \varepsilon$ expansion, with the observed sign, and is incompatible with current-algebra primaries of a WZW model.
4. *Gauss–Bonnet cosmic branes* (Section 5). In a companion paper [25] we applied the cosmic-brane description to fractal entangling surfaces. There, in Einstein gravity, the shape of $f(\alpha)$ is fixed by the bulk through the horizon radius x_q , and in $d = 2$ it reduces to the Calabrese–Lefevre form, which is universal for all CFTs. Here we extend the analysis to Gauss–Bonnet gravity. The q -expansion of the holographic mass exponent is controlled by CFT data such as C_T/a . Positive Gauss–Bonnet coupling leads to a negative refined Rényi entropy at finite q . No dictionary between holographic couplings and b_1 is warranted.

Section 6 discusses limitations, and the Appendix collects numerical methods and large-deviation derivations.

2 Entanglement thermodynamics and the ideal Fermi-gas theorem

2.1 The thermodynamic dictionary

Let $\mathcal{L} \rightarrow \infty$ be a large parameter, for instance $\mathcal{L} = \ln(1/\varepsilon)$ or $\mathcal{L} = \ln L$, and suppose

$$\ln \text{Tr} \rho_A^q = -\tau(q) \mathcal{L} + o(\mathcal{L}) \quad (1)$$

on an open interval I of q on which τ is differentiable.

Proposition 2.1 (Thermodynamic dictionary). *Under (1), for $q \in I$,*

$$S_q = D_q \mathcal{L} + o(\mathcal{L}), \quad D_q = \frac{\tau(q)}{q-1}, \quad (2)$$

$$\tilde{S}_q \equiv q^2 \partial_q \left(\frac{q-1}{q} S_q \right) = f(\alpha_q) \mathcal{L} + o(\mathcal{L}), \quad f(\alpha_q) = q\alpha_q - \tau(q), \quad \alpha_q = \tau'(q), \quad (3)$$

and $\tilde{S}_q$ is the von Neumann entropy of the escort state $\rho_A^q / \text{Tr} \rho_A^q$.

Proof. $\Phi(q) = \ln \text{Tr} \rho_A^q$ is convex, being a log-sum-exp of functions linear in q . Pointwise convergence of the convex functions $\Phi/\mathcal{L}$ to a limit that is differentiable on I implies convergence of their derivatives [7, Thm. 25.7]. Since $\frac{q-1}{q} S_q = -\Phi/q$, we have $\tilde{S}_q = \Phi - q\Phi'$, which converges to $(q\tau' - \tau)\mathcal{L}$. For the escort state $\pi_i = \lambda_i^q / \sum_j \lambda_j^q$, one finds $-\sum_i \pi_i \ln \pi_i = q\langle -\ln \lambda \rangle_\pi + \Phi = \Phi - q\Phi'$. $\square$

Particular values have direct information-theoretic meaning: D_0 is the Hartley entropy (log-rank), D_1 the von Neumann entropy, D_2 the collision entropy, and $D_\infty = \alpha_{\min}$ the min-entropy, all per unit $\mathcal{L}$. Figure 1 illustrates the dictionary for a two-scale Cantor measure read as an entanglement spectrum. There ρ_A is either the exact cascade spectrum at finite resolution, or a tensor product of n “bond” qubits with Schmidt weights (p_1, p_2) . In the second case $S_q = n H_q(p)$ exactly, so $D_q = H_q(p)/\ln 2$.

2.2 The ideal Fermi-gas theorem

For a Gaussian fermionic state, Peschel’s theorem [9] gives $\rho_A = Z^{-1} \exp(-\sum_k \varepsilon_k c_k^\dagger c_k)$. The entanglement energies ε_k are related to the eigenvalues ζ_k of the correlation matrix $C_{ij} = \langle c_i^\dagger c_j \rangle$, $i, j \in A$, by $\zeta_k = (1 + e^{\varepsilon_k})^{-1}$.

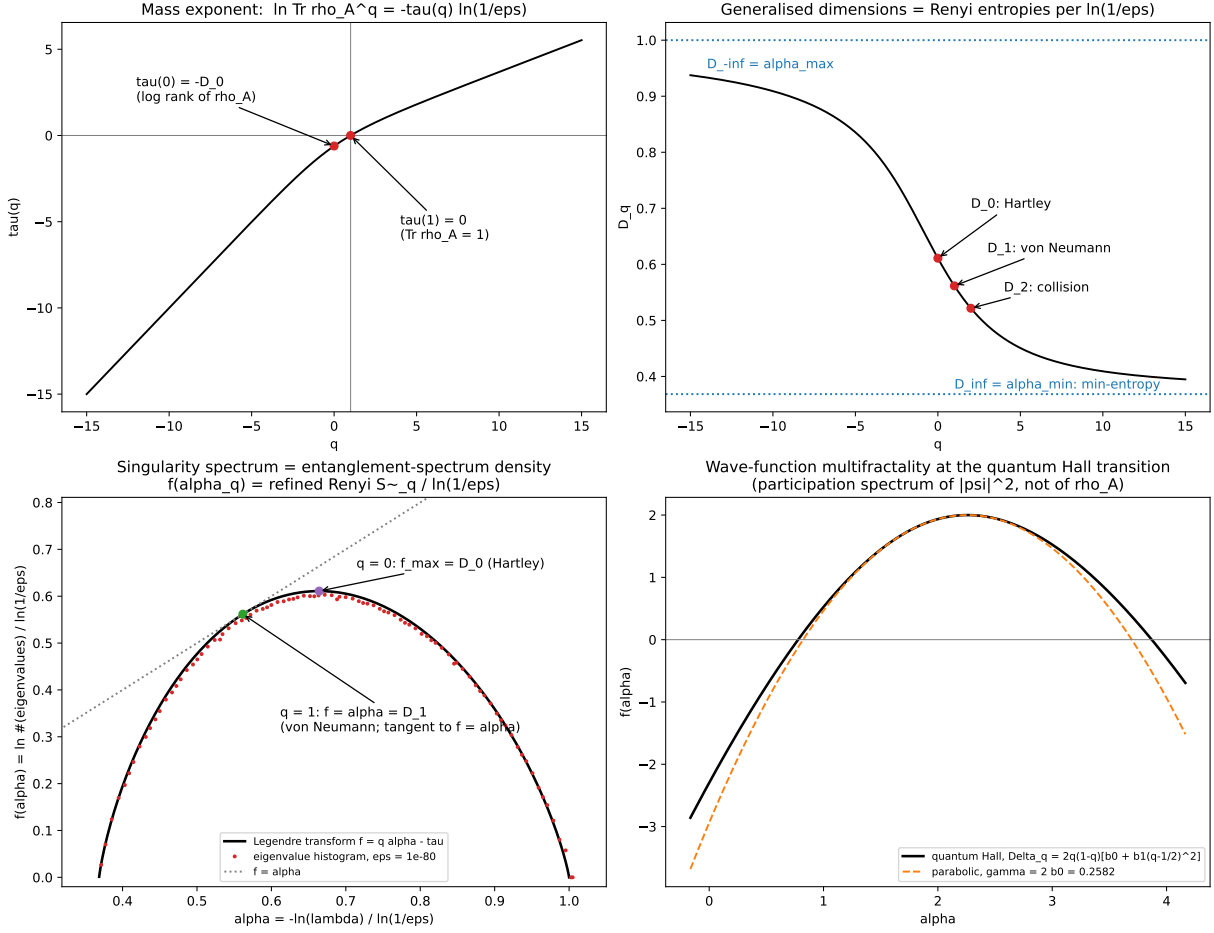


Figure 1: Thermodynamic dictionary for a two-scale Cantor measure ($p = (0.6, 0.4)$, $r = (0.25, 0.4)$) read as the eigenvalue distribution of ρ_A . Top left: mass exponent, with $\tau(0) = -D_0$ (log-rank) and $\tau(1) = 0$ (normalisation). Top right: generalised dimensions $D_q = S_q/\mathcal{L}$, with the Hartley, von Neumann, collision and min-entropies marked. Bottom left: singularity spectrum from the Legendre transform (line) and from the exact eigenvalue histogram at $\varepsilon = 10^{-80}$ (dots); the von Neumann point is tangent to $f = \alpha$. Bottom right, for contrast: the quantum Hall wave-function spectrum $\Delta_q = 2q(1-q)[b_0 + b_1(q - \frac{1}{2})^2]$ of [8] and its parabolic approximation. This is a participation spectrum of $|\psi|^2$, not an entanglement spectrum.

Theorem 1 (Ideal Fermi-gas theorem). *Let ρ_A be Gaussian with entanglement energies $\{\varepsilon_k\}$, and set $n_q(\varepsilon) = (1 + e^{q\varepsilon})^{-1}$ and $s_F(n) = -n \ln n - (1 - n) \ln(1 - n)$. Then, exactly,*

$$\ln \text{Tr} \rho_A^q = \sum_k \left[\ln(1 + e^{q\varepsilon_k}) - q \ln(1 + e^{\varepsilon_k}) \right], \quad (4)$$

$$\tilde{S}_q = \sum_k s_F(n_q(\varepsilon_k)), \quad (5)$$

$$\langle K \rangle_q = \sum_k \left[\ln(1 + e^{\varepsilon_k}) - \varepsilon_k(1 - n_q(\varepsilon_k)) \right]. \quad (6)$$

In words, $\tilde{S}_q$ is the entropy of an ideal Fermi gas with single-particle levels ε_k at temperature $1/q$. If $\rho_A(\varepsilon) = \mathcal{L}^{-1} \sum_k \delta(\varepsilon - \varepsilon_k)$ has a limit, then $f_{\text{ent}}(\alpha_q) = \int d\varepsilon \rho_A(\varepsilon) s_F(n_q(\varepsilon))$.

Proof. ρ_A factorises over modes, with mode k having eigenvalues ζ_k and $1 - \zeta_k$, so $\text{Tr} \rho_A^q = \prod_k [\zeta_k^q + (1 - \zeta_k)^q]$. With $\zeta = (1 + e^\varepsilon)^{-1}$ one has $\zeta^q + (1 - \zeta)^q = (1 + e^{q\varepsilon})/(1 + e^\varepsilon)^q$, which gives (4). The escort state $\rho_A^q / \text{Tr} \rho_A^q \propto \exp(-q \sum_k \varepsilon_k c_k^\dagger c_k)$ is Gaussian with occupations $n_q(\varepsilon_k)$, and its

von Neumann entropy is $\sum_k s_F(n_q)$; by Proposition 2.1 this is $\tilde{S}_q$. Finally, $\langle K \rangle_q = -\partial_q \ln \text{Tr} \rho_A^q$, which gives (6). $\square$

Corollary 2.1. (i) $f_{\text{ent}} \geq 0$ and f_{ent} is concave, as is any microcanonical entropy of an ideal gas. (ii) f_{ent} depends on the state only through $\rho_A(\varepsilon)$. Each summand in (4) is exactly invariant under $\varepsilon \rightarrow -\varepsilon$ (particle-hole exchange of a mode), so f_{ent} depends only on the distribution of $|\varepsilon|$. (iii) A uniform density $\rho_A(\varepsilon) = 1/\pi^2$ per unit $\mathcal{L} = \ln \ell$ gives, using $\int_{\mathbb{R}} s_F((1+e^x)^{-1}) dx = \pi^2/3$,

$$f_{\text{ent}}(\alpha_q) = \frac{1}{3q}, \quad (7)$$

which is the Calabrese–Lefevre result $f(\alpha_q) = c/(3q)$ with $c = 1$ [4].

Remark (Bosonic Gaussian states). The same transcription holds for bosonic Gaussian states. By Williamson’s theorem the reduced state is $\rho_A = \bigotimes_k (1 - e^{-\varepsilon_k}) e^{-\varepsilon_k b_k^\dagger b_k}$, with symplectic eigenvalues $\nu_k = \frac{1}{2} \coth(\varepsilon_k/2)$ (vacuum normalised to $\nu = \frac{1}{2}$). Then

$$\ln \text{Tr} \rho_A^q = \sum_k \left[q \ln(1 - e^{-\varepsilon_k}) - \ln(1 - e^{-q\varepsilon_k}) \right], \quad \tilde{S}_q = \sum_k s_B(\bar{n}_q(\varepsilon_k)), \quad (8)$$

with $\bar{n}_q(\varepsilon) = (e^{q\varepsilon} - 1)^{-1}$ and $s_B(n) = (1+n) \ln(1+n) - n \ln n$. The refined Rényi entropy is thus the entropy of an ideal Bose gas at temperature $1/q$, by the same argument as in Theorem 1: the escort state is thermal with energies $q\varepsilon_k$. Pure modes ($\nu_k = \frac{1}{2}$) do not contribute, and $f_{\text{ent}} \geq 0$ is again concave. Unlike the fermionic case there is no $\varepsilon \rightarrow -\varepsilon$ symmetry, since $\varepsilon_k > 0$.

2.3 What wave-function multifractality can and cannot determine

The correlation matrix of a Slater determinant is $C = P_A \Pi P_A$, where $\Pi = \sum_{n \in \text{occ}} |\psi_n\rangle \langle \psi_n|$ and P_A projects onto A . Its non-trivial spectrum coincides with that of the overlap matrix $O_{nm} = \langle \psi_n | P_A | \psi_m \rangle$ restricted to occupied states.

Proposition 2.2 (Structure of the map from wave functions to entanglement). (i) f_{ent} is a functional of the spectrum of O , i.e. of joint statistics of overlaps between different eigenstates. Single-state moments $\langle |\psi_n|^{2q} \rangle$, and hence f_{wave} , do not determine it. For example, the particle-number variance $\kappa_2 = \text{Tr} O(1 - O) = \sum_{n \in \text{occ}, m \notin \text{occ}} |\langle \psi_n | P_A | \psi_m \rangle|^2$, which controls the leading entanglement [10], involves cross-energy correlations.

(ii) There is no rescaling under which f_{ent} coincides with a disorder-averaged f_{wave} on its full support. The latter can be negative and obeys $\Delta_q = \Delta_{1-q}$ [11], while $f_{\text{ent}} \geq 0$ (Corollary 2.1) and τ_{ent} has no $q \rightarrow 1 - q$ symmetry (for instance, $\tau \propto q - 1/q$ for a CFT).

Proof. (i) follows from $\text{spec}(P_A \Pi P_A) \setminus \{0\} = \text{spec}(\Pi P_A \Pi) \setminus \{0\}$ and Theorem 1. (ii) follows from the stated properties. $\square$

Numerical test. We diagonalised critical power-law random banded matrices (PRBM) [12] of the unitary class with $b = 1$ and $N = 2048$ (six realisations). We filled the lower half of the spectrum and took A to be half of the chain. The identity (5) holds to a relative accuracy of 5×10^{-10} . The wave-function spectrum of states near the band centre gives $D_2 \simeq 0.81$ and $\alpha_0 \simeq 1.11$ at this size. The entanglement spectrum, normalised by $S_1 = 13.7$, is non-negative everywhere, with minimum 0.026, and departs visibly from the clean Calabrese–Lefevre form (Fig. 2, top left). Disorder thus changes f_{ent} , but only through $\rho_A(\varepsilon)$, as Theorem 1 requires.

3 Markov random cascades in tensor networks

3.1 From random tensor networks to cascades

If a state can be written as $(V_A \otimes V_B) \bigotimes_{e \in \gamma} |\phi_e\rangle$ with isometries V_A, V_B and bond states $|\phi_e\rangle$ on a cut γ , then the non-zero spectrum of ρ_A is that of $\bigotimes_e \text{Tr}_e |\phi_e\rangle \langle \phi_e|$. For random tensor networks with large bond dimension, Rényi entropies are given by sums of bond Rényi entropies along the minimal cut [13]. For a MERA-like network [14], the cut of an interval crosses a bounded number of bonds per layer over $n \propto \log_b L$ layers. With identical bond spectra $\{p_j\}$, the spectrum of ρ_A is then a multinomial cascade, with

$$\tau(q) = -\frac{1}{\ln b} \ln \sum_j p_j^q \quad (\text{per layer, } \mathcal{L} = \ln L). \quad (9)$$

The continuum $f(\alpha)$ arises from the discrete layers through Cramér's theorem: $\ln \lambda$ is a sum of n i.i.d. terms, and f is the Legendre transform of their cumulant generating function (Appendix B). For a genuine MERA, disentanglers spoil the exact product structure, and the cascade is a model.

3.2 Markov modulation

To model residual correlations between layers, let the bond spectrum of layer k depend on a sector s_k that follows a Markov chain with transition matrix T and stationary distribution π . Define $z_q(s) = \sum_j p_{sj}^q$ and $D_q = \text{diag}(z_q)$. The annealed partition function over n layers is $Z_n(q) = \pi^\top D_q (TD_q)^{n-1} \mathbf{1}$.

Proposition 3.1 (Perron-root mass exponent). *For irreducible aperiodic T , $\tau(q) = -\ln \Lambda_q / \ln b$, where Λ_q is the Perron root of TD_q . Λ_q is real-analytic in q , so τ and f have no singularities at finite q .*

Proof. TD_q is non-negative, irreducible and aperiodic, so its Perron root is simple and strictly dominant, and $\frac{1}{n} \ln Z_n \rightarrow \ln \Lambda_q$. The matrix entries are analytic in q , and a simple isolated eigenvalue of an analytic matrix family is analytic [15]. $\square$

Proposition 3.2 (Two-state chain). *For the symmetric two-state chain with second eigenvalue λ_2 and $\pi = (\frac{1}{2}, \frac{1}{2})$, writing $S = z_1 + z_2$ and $P = z_1 z_2$,*

$$\Lambda_q = \frac{1}{4} \left[(1 + \lambda_2) S + \sqrt{(1 + \lambda_2)^2 S^2 - 16 \lambda_2 P} \right], \quad (10)$$

$$\Delta\tau(q) \equiv \tau(q) - \tau(q)|_{\lambda_2=0} = -\frac{\lambda_2}{\ln b} \frac{\text{Var}_\pi(z_q)}{\langle z_q \rangle_\pi^2} + O(\lambda_2^2). \quad (11)$$

Proof. With $T = \begin{pmatrix} 1-e & e \\ e & 1-e \end{pmatrix}$ and $\lambda_2 = 1 - 2e$, the matrix TD_q has trace $\frac{1+\lambda_2}{2} S$ and determinant $\lambda_2 P$, which gives the first formula. Differentiating at $\lambda_2 = 0$ gives $\partial_{\lambda_2} \Lambda_q = (z_1 - z_2)^2 / (2S) = \text{Var}_\pi(z_q) / \langle z_q \rangle_\pi$. $\square$

Two features distinguish (11) from a naive guess $\Delta\tau \propto \lambda_2^q$. The correction is *linear* in λ_2 , because λ_2 weights a single transition per layer irrespective of q , and q enters only through $\text{Var}_\pi(z_q)$. Moreover $z_1(1) = z_2(1) = 1$, so $\Delta\tau(1) = 0$ and normalisation is preserved.

Phase transitions. By Proposition 3.1 there are none for $\lambda_2 < 1$. At $\lambda_2 = 1$ the chain is reducible and $\Lambda_q = \max(z_1, z_2)$. This has a kink wherever $z_1(q) = z_2(q)$ with different slopes, and in particular at $q = 1$ whenever the two sectors have different von Neumann entropies. The Legendre transform then contains a linear segment, i.e. a Maxwell construction and a first-order transition. For $\lambda_2 \rightarrow 1$ the discriminant shows that the crossover sharpens on the scale $|z_1 - z_2| \sim (1 - \lambda_2)^{1/2} S$. Because the partition function is annealed over sectors, f may become negative, by the same mechanism that makes disorder-averaged wave-function spectra negative. A single ρ_A always has $f \geq 0$.

Numerical test. With bond spectra $(0.7, 0.3)$ and $(0.9, 0.1)$ and $b = 2$, the first-order formula reproduces the exact Perron result with relative error 2.1×10^{-2} at $\lambda_2 = 0.05$ and 8.4×10^{-2} at $\lambda_2 = 0.2$, i.e. with error $O(\lambda_2^2)$. A finite transfer-matrix product with $n = 400$ layers gives 0.6197 against the Perron value 0.6194 at $q = 2.5$, $\lambda_2 = 0.5$. The sharpening of the crossover as $\lambda_2 \rightarrow 1$ is shown in Fig. 2 (top right).

4 Non-parabolicity and Casimir invariants at the quantum Hall critical point

4.1 Weyl invariance and the Casimir variable

At Anderson transitions the multifractal exponents obey $\Delta_q = \Delta_{1-q}$ [11]. Gruzberg, Ludwig, Mirlin and Zirnbauer traced this symmetry to the invariance of the nonlinear σ -model under the Weyl group of its target space [16]. Scaling operators are labelled by highest weights, and at one loop their dimensions are proportional to the quadratic Casimir eigenvalue [17]. For the operators $|\psi|^{2q}$ the Weyl reflection acts as $q \rightarrow 1 - q$. Any Weyl-invariant function of q is therefore a function of

$$u = q(1 - q). \quad (12)$$

Proposition 4.1 (Casimir form of the quantum Hall spectrum). *The high-precision quantum Hall result [8], $\Delta_q = 2q(1 - q)[b_0 + b_1(q - \frac{1}{2})^2]$ with $b_0 = 0.1291 \pm 0.0002$ and $b_1 = 0.0029 \pm 0.0003$, is identical to*

$$\Delta_q = c_1 u + c_2 u^2, \quad c_1 = 2(b_0 + \frac{b_1}{4}) = 0.2596, \quad c_2 = -2b_1 = -0.0058. \quad (13)$$

In particular $\alpha_0 - 2 = \Delta'_0 = c_1$, in agreement with the reported $\alpha_0 = 2.2596$.

Proof. Use $(q - \frac{1}{2})^2 = \frac{1}{4} - u$, which gives $2u[b_0 + b_1(\frac{1}{4} - u)] = 2(b_0 + \frac{b_1}{4})u - 2b_1 u^2$. Also $\Delta'_0 = c_1$, since $u'(0) = 1$ and $u(0) = 0$. $\square$

A parabolic spectrum is thus exactly the statement that Δ_q is linear in the quadratic Casimir. The non-parabolicity b_1 measures the u^2 term, i.e. the contribution of higher Casimir invariants.

4.2 Origin in the loop expansion

In $2 + \varepsilon$ dimensions the unitary-class result through four loops is [3]

$$\Delta_q = q(1 - q) \left(\frac{\varepsilon}{2} \right)^{1/2} - \frac{3}{8} \zeta(3) q^2 (q - 1)^2 \varepsilon^2 + O(\varepsilon^{5/2}) = u \left(\frac{\varepsilon}{2} \right)^{1/2} - \frac{3}{8} \zeta(3) u^2 \varepsilon^2 + \dots \quad (14)$$

The u^2 term first appears at four loops, through the $\zeta(3)$ contribution. It gives $c_2 = -\frac{3}{8} \zeta(3) \varepsilon^2 < 0$, that is,

$$b_1 = \frac{3}{16} \zeta(3) \varepsilon^2 > 0, \quad (15)$$

which has the same sign as the measured value. We stress that the $2 + \varepsilon$ expansion is not controlled at the IQH fixed point, which lies at strong coupling with topological angle $\theta = \pi$. The loop expansion therefore explains the *type* and *sign* of the non-parabolic term, but not its magnitude.

4.3 Incompatibility with current-algebra primaries

Proposition 4.2. *If the operators $|\psi|^{2q}$ at the IQH fixed point were primaries of an affine current algebra realised through the Sugawara construction (as in a WZW model), their dimensions would be proportional to the quadratic Casimir, and b_1 would vanish.*

Proof. For current-algebra primaries in a representation R , the Sugawara construction gives $h_R = C_2(R)/(k + h^\vee)$ [18]. The dimension is then a linear function of C_2 , hence of u , and $c_2 = 0$. $\square$

Since $b_1 = 0.0029 \pm 0.0003$ differs from zero by about ten standard deviations [8], theories in which the multifractal operators are Sugawara primaries are excluded. This does not exclude a WZW model perturbed by an operator that breaks the current algebra, nor a description in which $|\psi|^{2q}$ are not current-algebra primaries. We do not know which operator generates the u^2 term at the IQH fixed point.

5 Holographic duals and higher-curvature constraints

5.1 Cosmic branes and the two-dimensional case

In holographic CFTs the refined Rényi entropy is the area of a cosmic brane of tension $T_q = (q - 1)/(4qG_N)$ [6]. For spherical regions the brane is the horizon of a hyperbolic black hole at temperature T_0/q [19]. In $d = 2$ the q -dependence of Rényi entropies is universal for every CFT, $S_q = \frac{c}{6}(1 + \frac{1}{q})\ln(\ell/\varepsilon)$. The corresponding spectrum is $f(\alpha) = \sqrt{\frac{c}{3}(2\alpha - \frac{c}{3})}$ [4], which is of square-root type rather than parabolic and has no maximum: $\alpha \rightarrow \infty$ as $q \rightarrow 0$. In a three-dimensional bulk the Gauss–Bonnet term vanishes identically, so higher-curvature terms of this type cannot change the two-dimensional result.

5.2 Gauss–Bonnet branes in $d = 4$

For Gauss–Bonnet gravity in $d + 1 = 5$ dimensions, define $a = \lambda_{\text{GB}} f_\infty = \frac{1}{2}(1 - \sqrt{1 - 4\lambda_{\text{GB}}})$. The temperature and horizon entropy of the hyperbolic black hole with horizon radius x , in units of the AdS radius [20], are

$$\frac{T(x)}{T_0} = \frac{d(1 - a)x^4 - (d - 2)x^2 + (d - 4)a}{2x(x^2 - 2a)}, \quad \frac{S(x)}{S_1} = x^{d-1} \frac{1 - \frac{2a(d-1)}{(d-3)x^2}}{1 - \frac{2a(d-1)}{d-3}}. \quad (16)$$

Here $S(x)$ is the Jacobson–Myers horizon entropy, which replaces the area in Dong’s prescription for this theory [21]. The brane sits at x_q , defined by $T(x_q)/T_0 = 1/q$. Define $I(q) = \int_{x_q}^1 \frac{S}{S_1} d(\frac{T}{T_0})$. The thermodynamic method of [19] then gives, exactly,

$$\tau(q) = q I(q), \quad \alpha_q = I(q) + \frac{S(x_q)}{q S_1}, \quad f(\alpha_q) = \frac{S(x_q)}{S_1}, \quad (17)$$

in units of S_1 . The last relation is Dong’s formula, which is automatic in this construction.

C_T universality. Writing $\tau(q) = (q - 1) + t_2(q - 1)^2 + \dots$, Perlmutter’s theorem [22] fixes t_2 in terms of C_T relative to the normalisation of S_1 . For Gauss–Bonnet gravity in $d = 4$ one has $C_T \propto 1 - 2a$ and $a_{\text{anom}} \propto 1 - 6a$ [23], so $t_2^{\text{GB}}/t_2^{\text{E}} = (1 - 2a)/(1 - 6a)$. Table 1 confirms this numerically to about 10^{-3} . In the Einstein limit, the numerically integrated S_q agrees with the closed form of [19] to 3×10^{-10} .

λ_{GB}	a	t_2	t_2/t_2^{E}	$(1-2a)/(1-6a)$
$-7/36$	-0.1667	-0.3303	0.6672	0.6667
-0.10	-0.0916	-0.3782	0.7639	0.7635
0	0	-0.4951	1	1
0.05	0.0528	-0.6479	1.3085	1.3090
0.09	0.1000	-0.9895	1.9985	2.0000

Table 1: Second Taylor coefficient of the holographic mass exponent for Gauss–Bonnet gravity in $d = 4$, over the allowed range $-7/36 \leq \lambda_{\text{GB}} \leq 9/100$.

Negative refined Rényi entropy at finite q . For $a > 0$ the Jacobson–Myers entropy changes sign at $x^2 = 2a(d-1)/(d-3)$. As $q \rightarrow \infty$, $x_q^2 \rightarrow (d-2)/(d(1-a))$ for $d = 4$. At $\lambda_{\text{GB}} = 0.09$ we find that $f(\alpha_q) = \tilde{S}_q/S_1$ becomes negative for $q > q_c \simeq 6.47$ (Fig. 2, bottom right). Because $\tilde{S}_q$ is the entropy of a density matrix, it cannot be negative. The hyperbolic black-hole saddle therefore cannot dominate beyond q_c , and another saddle, or a breakdown of the effective description, must take over. This is a saddle-exchange transition of the kind discussed in Section 3.

5.3 No dictionary between b_1 and λ_{GB}

It is tempting to relate the quantum Hall coefficient b_1 to a holographic coupling such as λ_{GB} . We argue that this would be unfounded, for three reasons.

1. The measures differ: a disorder-averaged participation spectrum (f may be negative, $\Delta_q = \Delta_{1-q}$) versus the entanglement spectrum of a clean, strongly coupled CFT ($f \geq 0$, no $q \rightarrow 1-q$ symmetry).
2. The dimensions differ: the IQH transition is two-dimensional, where holographic Rényi entropies are universal, whereas Gauss–Bonnet effects require $d \geq 4$.
3. The Weyl-invariant variable u that organises the quantum Hall spectrum has no counterpart in τ_{ent} .

What the two problems share is an organising principle. In each case the departure from the leading form is controlled by the next invariant of the critical theory: the u^2 (higher-Casimir) term for localisation, and CFT data such as C_T/a and $\langle TTT \rangle$ for holography.

6 Discussion and limitations

1. Theorem 1 and the subsequent Remark are exact for fermionic and bosonic Gaussian states, respectively. For interacting systems the modular Hamiltonian is not quadratic, and f_{ent} is no longer an ideal-gas entropy.
2. Proposition 2.2 is a structural statement. We have not derived $\rho_A(\varepsilon)$ from the multi-state correlations of critical wave functions; doing so would require the joint statistics of overlaps across energies, beyond the single-state exponents Δ_q .
3. The PRBM numerics use a single system size, so the extracted exponents ($D_2 \simeq 0.81$, $\alpha_0 \simeq 1.11$) carry finite-size corrections that we have not quantified.
4. The Markov cascade is a phenomenological model of residual inter-layer correlations in tensor networks, and its partition function is annealed. Quenched (typical) averages may exhibit freezing-type transitions that we have not analysed.

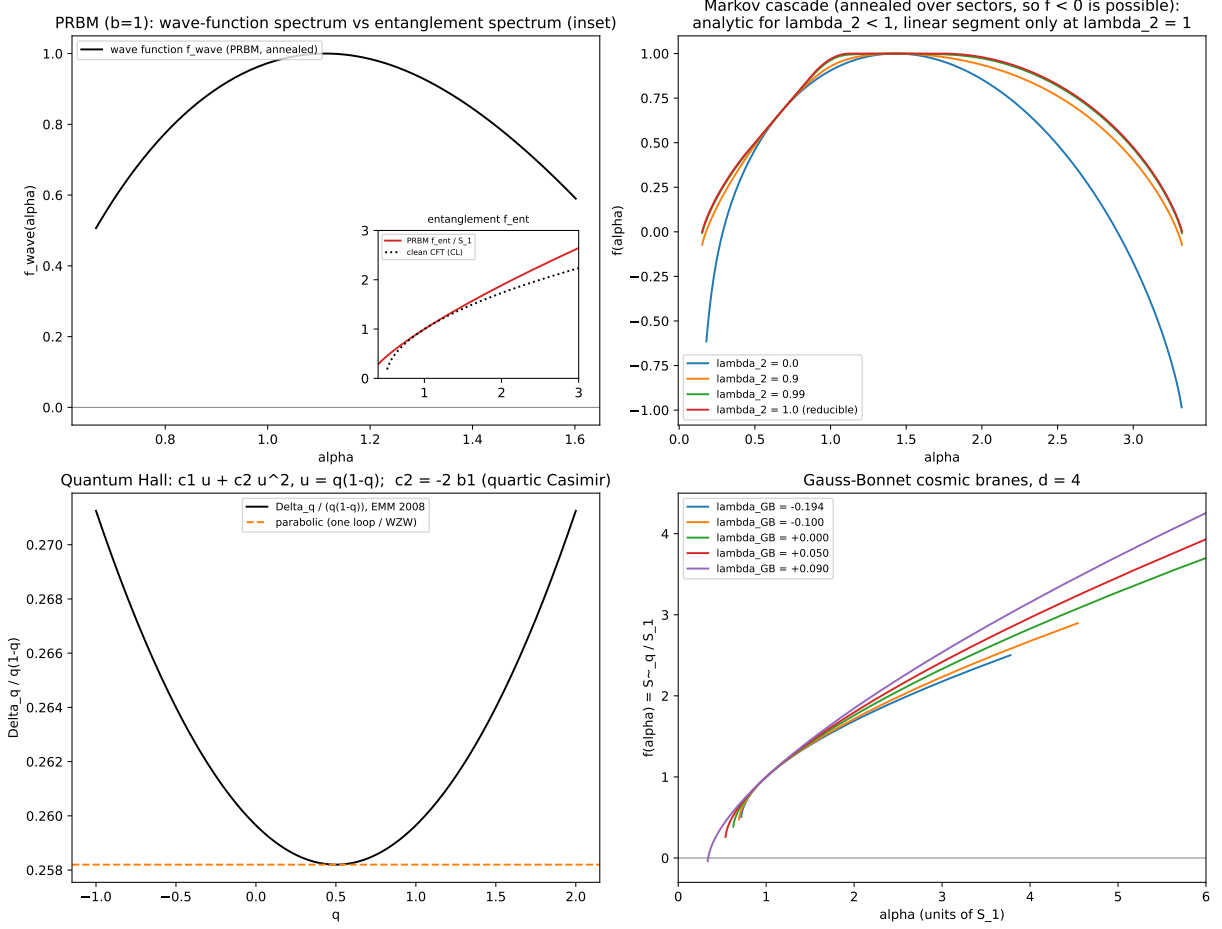


Figure 2: Top left: wave-function singularity spectrum of a critical PRBM ($b = 1$, $N = 2048$, annealed over states near the band centre); inset: entanglement spectrum f_{ent}/S_1 of the half-filled Slater determinant compared with the clean Calabrese–Lefevre form. Top right: Markov-modulated cascade, annealed over sectors, for $\lambda_2 = 0, 0.9, 0.99, 1$; the spectrum is analytic for $\lambda_2 < 1$ and develops a linear segment at $\lambda_2 = 1$. Bottom left: quantum Hall $\Delta_q/q(1-q)$ [8] against the parabolic value; the curvature is the u^2 coefficient $c_2 = -2b_1$. Bottom right: Gauss–Bonnet cosmic-brane spectra in $d = 4$.

5. Section 4 identifies the type and sign of the non-parabolic term, but the $2 + \varepsilon$ expansion is not controlled at the IQH fixed point. The operator generating the u^2 term remains unknown.
6. The negative refined Rényi entropy for $\lambda_{\text{GB}} > 0$ indicates a change of dominant saddle; we have not constructed the competing saddle.
7. The singularity spectra studied here are Shannon–Rényi (statistical) measures of the entanglement spectrum. They coincide with algorithmic (Kolmogorov) complexity only on average, up to the complexity of the underlying distribution [26]. How algorithmic complexity scales with resolution, and where uncomputability enters, is analysed in a companion paper [27].

Data and code availability

All figures and numbers are produced by two Python scripts, `multifractal_entanglement.py` (Section 2, Fig. 1) and `frontier2_deepdive.py` (Sections 2–5, Fig. 2, Table 1), available

A Numerical methods

- *Two-scale Cantor measure.* $\tau(q)$ is obtained from $\sum_i p_i^q r_i^{-\tau} = 1$ by Brent’s method, with $\alpha = \sum_i w_i \ln p_i / \sum_i w_i \ln r_i$ and $w_i = p_i^q r_i^{-\tau}$. The exact spectrum at resolution ε is enumerated by composition (k_1, k_2) with the stopping rule “first prefix of size $\leq \varepsilon$ ”, using logarithmic binomial coefficients. The trace equals 1 to 10^{-12} . At $\varepsilon = 10^{-80}$, $S_q/\mathcal{L}$ differs from D_q by $O(1/\mathcal{L})$, e.g. 0.5633 against 0.5616 at $q = 1$.
- *PRBM.* H_{ij} is complex Gaussian with variance $[1 + (r_{ij}/b)^2]^{-1}$, using the periodic distance r_{ij} , and is Hermitised. C is formed from the lower half of the eigenvectors. Wave-function moments use box sizes $2, 4, \dots, N/8$ and the 64 states closest to the band centre. Entanglement quantities use $\ln(\zeta^q + (1 - \zeta)^q)$ with ζ clipped to $[10^{-15}, 1 - 10^{-15}]$.
- *Markov cascade.* The Perron root is computed by dense diagonalisation of TD_q . The finite- n check multiplies normalised row vectors for $n = 400$ layers.
- *Gauss–Bonnet.* x_q is found by Brent’s method above the largest zero of the numerator of $T(x)$. $I(q)$ is computed by adaptive quadrature, and α_q from the exact relation in Section 5. t_2 is obtained by central differences on a uniform grid in q .

B Large-deviation derivations

Equal-scale cascade (Cramér). With n independent layers and bond spectrum $\{p_j\}$, the eigenvalues are $\lambda = \prod_k p_{j_k}$. Their number with $-\ln \lambda / (n \ln b) \in [\alpha, \alpha + d\alpha]$ is a multinomial count. By Stirling’s formula, or equivalently Cramér’s theorem for the empirical frequencies [24], it behaves as $b^{nf(\alpha)}$, with f the Legendre transform of $\tau(q) = -\log_b \sum_j p_j^q$.

Markov modulation (Gärtner–Ellis). For an irreducible Markov additive process the scaled cumulant generating function exists and equals $\ln \Lambda_q$, the logarithm of the Perron root of the tilted matrix [24]. Since it is differentiable, the Gärtner–Ellis theorem yields the large-deviation rate function as its Legendre transform, which is f up to normalisation.

Crossover width near $\lambda_2 = 1$. For $\lambda_2 = 1 - 2e$ the discriminant in Proposition 3.2 is $4(z_1 - z_2)^2 + O(e)S^2$. The smooth Λ_q therefore deviates from $\max(z_1, z_2)$ appreciably only where $(z_1 - z_2)^2 \lesssim eS^2$, i.e. on a window of width $\propto (1 - \lambda_2)^{1/2}$ around each crossing.

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